

DAY-8

- ❖ **VLAN Trunking and Access Ports Theory : Trunking protocols (802.1Q, ISL). Practical: Configure trunk ports on a switch. Task: Simulate communication between VLANs using trunk ports.**

1. In Cisco Packet Tracer, configure two switches (Switch1 and Switch2) and connect them with a crossover cable. Set the FastEthernet0/1 port on both switches as trunk ports using the 802.1Q protocol.

ANS :

Objective

Connect **Switch1** and **Switch2** and configure FastEthernet0/1 on both switches as trunk ports using **802.1Q**.

Topology

Switch1 Fa0/1 ===== Fa0/1 Switch2

Trunk Link

802.1Q

Use a **crossover cable** between:

Switch1 Fa0/1

|
|

Switch2 Fa0/1

Switch1 Configuration

enable
configure terminal

interface fastethernet0/1
switchport mode trunk
no shutdown
exit

end

Switch2 Configuration

enable
configure terminal

interface fastethernet0/1
switchport mode trunk
no shutdown
exit

end

Verify the trunk

On both switches:
show interfaces trunk
You should see **Fa0/1** listed as a trunk port.

Result

Fa0/1 = Trunk
Encapsulation = 802.1Q
Status = Trunking

Conclusion:

The link between Switch1 and Switch2 is configured as a trunk, allowing multiple VLANs to travel through the same physical connection.

2. Create three VLANs (10, 20, 30) on both switches and assign FastEthernet0/2, 0/3, and 0/4 on Switch1 to VLAN 10, 20, and 30 respectively as access ports.

ANS:

Objective

Create VLANs:
VLAN 10
VLAN 20
VLAN 30

Assign the following ports on **Switch1**:

Switch1 Port	VLAN	Port Type
Fa0/2	VLAN 10	Access
Fa0/3	VLAN 20	Access
Fa0/4	VLAN 30	Access

Switch1 Configuration

enable
configure terminal

vlan 10
name VLAN10
exit

vlan 20
name VLAN20
exit

vlan 30
name VLAN30

exit

```
interface fastethernet0/2
switchport mode access
switchport access vlan 10
no shutdown
exit
```

```
interface fastethernet0/3
switchport mode access
switchport access vlan 20
no shutdown
exit
```

```
interface fastethernet0/4
switchport mode access
switchport access vlan 30
no shutdown
exit
```

end

Switch2 Configuration

Create the same VLANs on Switch2:

```
enable
configure terminal
```

```
vlan 10
name VLAN10
exit
```

```
vlan 20
name VLAN20
exit
```

```
vlan 30
name VLAN30
exit
```

end

Verify VLANs

```
show vlan brief
You should see:
```

```
10  VLAN10
20  VLAN20
30  VLAN30
```

Expected Result

Fa0/2 → VLAN 10

Fa0/3 → VLAN 20

Fa0/4 → VLAN 30

3. Connect three PCs to Switch1 on ports assigned to VLAN 10, 20, and 30. Verify that PCs in different VLANs cannot communicate with each other, but can communicate with other PCs in the same VLAN.

ANS :

Connect PCs

Connect three PCs to Switch1:

PC1 → Switch1 Fa0/2 → VLAN 10

PC2 → Switch1 Fa0/3 → VLAN 20

PC3 → Switch1 Fa0/4 → VLAN 30

Assign IP addresses.

PC1 - VLAN 10

IP Address: 192.168.10.10

Subnet Mask: 255.255.255.0

PC2 - VLAN 20

IP Address: 192.168.20.10

Subnet Mask: 255.255.255.0

PC3 - VLAN 30

IP Address: 192.168.30.10

Subnet Mask: 255.255.255.0

No default gateway is required for this test because we are testing communication **within and between VLANs without a router.**

Test 1: PC1 → PC2

On PC1:

ping 192.168.20.10

Result

Request timed out.

**4. Change the trunk encapsulation on the trunk link between Switch1 and Switch2 from 802.1Q to ISL, and test if VLAN traffic still passes between the switches.

Hint: Use the 'switchport trunk encapsulation isl' command before setting the port to trunk mode.**

ANS :

```
Switch1:
    enable
    configure terminal

    interface fastethernet0/1
    switchport trunk encapsulation ?
```

If you see:

```
dot1q
isl
```

You can configure:

```
interface fastethernet0/1
switchport trunk encapsulation isl
switchport mode trunk
no shutdown
exit
```

Switch2:

```
interface fastethernet0/1
switchport trunk encapsulation isl
switchport mode trunk
no shutdown
exit
```

Test the VLAN traffic

From a PC on Switch1, ping a PC belonging to the same VLAN on Switch2.

For example:

PC1 → PC5

where both PCs belong to VLAN 10.

Use:

ping 192.168.1.10

If ISL is supported

The VLAN traffic should continue to pass between the switches.

If ISL is NOT supported

Packet Tracer may show an error such as an invalid/unsupported encapsulation command. In that case, **you cannot perform the ISL test using that switch model.**

```
Keep the trunk as 802.1Q:  
interface fastethernet0/1  
switchport trunk encapsulation dot1q  
switchport mode trunk
```

Conclusion

802.1Q is the practical choice for modern switches. ISL is an older Cisco proprietary protocol and is not supported by many modern switch models.

5. Simulate a scenario where a new port on Switch2 (FastEthernet0/5) is accidentally set as a trunk port instead of an access port for VLAN 20. Observe and explain the impact on connectivity for a PC connected to that port.

ANS:

Objective

Suppose Switch2 Fa0/5 was supposed to be an **access port for VLAN 20**, but someone accidentally configured it as a trunk.

Correct configuration should have been:

```
interface fastethernet0/5  
switchport mode access  
switchport access vlan 20  
But accidentally we configure:  
interface fastethernet0/5  
switchport mode trunk  
no shutdown
```

Connect PC

Connect:

PC → Switch2 Fa0/5

The PC is now connected to a trunk port instead of a normal access port.

What happens?

A PC normally sends **untagged Ethernet frames**. A trunk is designed primarily to carry traffic for multiple VLANs between network devices.

Therefore, configuring an end-device port as a trunk can cause unexpected connectivity behavior and VLAN assignment problems.

The PC may **not behave as expected as a normal VLAN 20 device**.

Check the configuration

Run : show interfaces trunk

You may see: Fa0/5
listed as a trunk.

Also run:

show vlan brief

and:

show running-config interface fastethernet0/5

Correct the mistake

Change Fa0/5 back to an access port:

enable

configure terminal

interface fastethernet0/5

switchport mode access

switchport access vlan 20

no shutdown

exit

Verify:

show vlan brief

Fa0/5 should now appear under : VLAN 20

Test again

Give the PC a VLAN 20 IP address, for example:

IP Address: 192.168.20.20

Subnet Mask: 255.255.255.0

Then ping another VLAN 20 PC:

ping 192.168.20.10

The ping should succeed if the other PC is correctly configured.